ESP32-Based Battery Management System with Protection Relay, LCD Interface and Blynk IoT Telemetry

Project report | Platform: ESP32 DevKit-C V4 (Wokwi simulation) | Firmware: Arduino C++ | Prepared by: Mohammed Ayaan Farooqui| Date: 12 November 2026

1. Abstract

This project implements a four-cell battery monitoring and protection controller on an ESP32. Each cell voltage is sampled at 5 Hz, filtered, validated and analysed by a reusable template class (BmsEngine). A four-state protection state machine (NORMAL, WARNING, FAULT, RECOVERY) drives a protection relay, three status LEDs and a buzzer. A flicker-free 16x2 I2C LCD shows live status. A telemetry layer publishes 26 datastreams to a Blynk dashboard, buffers events in a 20-entry FIFO queue while the cloud is unreachable, and replays them in order after reconnection. A rule-based risk score and maintenance recommendation are computed on the device.

The design is simulated in Wokwi with four potentiometers standing in for cell-tap voltages. This report describes the architecture, hardware mapping, algorithms, test method and the accompanying test datasets, and records the issues found while reviewing the design.

2. Objectives

1. Measure up to N series cells (compile-time constant, 4 in this build) and derive pack SoC, imbalance, strongest/weakest cell and imbalance trend.
2. Reject bad data (noise, disconnected taps, frozen channels, unrealistic jumps) before it can trip or mask a protection decision.
3. Open the pack relay on a confirmed fault, and only re-close after a verified stable recovery dwell.
4. Keep protection logic independent of Wi-Fi and cloud availability.
5. Provide local (LCD, LEDs, buzzer) and remote (Blynk) visibility, with no loss of event ordering during outages.
6. Provide repeatable test datasets with expected outputs.

3. System Architecture

The firmware is a single cooperative loop with no RTOS tasks and no blocking delays in normal operation. Each pass of loop() services, in order: demo serial commands, the Wi-Fi/Blynk reconnect timers, Blynk.run(), network status reporting, the offline-queue replay, the paced telemetry transmitter, the buzzer, the recovery LED blink, the LCD engine, and finally (every 200 ms) the BMS sample and state machine.

Layer	Component	Responsibility
Sensing and analytics	`BmsEngine<NUM_CELLS>`	ADC read, median filter, sensor-health flags, SoC, imbalance, adaptive threshold, trend

Layer	Component	Responsibility
Protection	State machine in `loop()`	Debounced transitions, relay, LEDs, buzzer
Local UI	Differential LCD engine	Page rotation, fault override, per-character updates
Cloud	Event queue + paced transmitter	Change-driven snapshots, FIFO buffering, replay, Blynk events
Analytics	Risk score, fault history	Weighted score, 24 h fault count, recommendation text

The engine header has no knowledge of relays, LEDs, the LCD or the network. Scaling from 4 to 16 cells needs only a change to NUM_CELLS and the cell_Pins[] list.

4. Hardware Description (from diagram.json)

Function	Part	ESP32 pin	Notes
Cell 1 to 4 voltage	4 x potentiometer (SIG)	GPIO 32, 33, 34, 35	All on ADC1, so they remain usable while Wi-Fi is active
Green LED (NORMAL)	LED	GPIO 15	
Yellow LED (WARNING / recovery blink)	LED	GPIO 16	
Red LED (FAULT)	LED	GPIO 17	
Protection relay	Relay module (IN)	GPIO 2	Active-low in firmware (`RELAY_ACTIVE_HIGH false`)
Buzzer	Buzzer (pin 2)	**D3 in the diagram; GPIO 25 in the firmware**	See finding F1
LCD 1602	I2C backpack (addr 0x27)	SDA 21, SCL 22	Powered from 5V
Potentiometers, relay	VCC	3V3	GND shared on GND.2

The potentiometers span 0 to 3.3 V. In the firmware this range represents scaled cell voltages, with 3.20 V as the 100 % SoC and over-range reference.

5. Software Design

5.1 Sampling and filtering

Every 200 ms each channel is read with analogRead (12-bit, 0 to 4095 counts for 0 to 3.3 V). A five-sample median filter per channel removes isolated spikes: one or two outlier samples cannot pass through a window of five, but three consecutive outliers can. The consequence is a step-response lag of three samples (about 0.6 s) before a real change appears in the filtered voltage.

5.2 Sensor-health checks

Check	Rule (raw counts)	Result
Out of range	raw < 248 (0.20 V) or raw > 3970 (3.20 V)	Immediate sensor fault
Frozen	change of 1 count or less for 150 consecutive samples (30 s)	Sensor fault
Unrealistic jump	change of more than 800 counts (about 0.645 V) in one sample, confirmed on 3 consecutive samples	Sensor fault
Correlated transient	jumps on at least 2 of 4 channels in the same sample	Treated as a load transient, jump counters reset, no fault

The correlation rule distinguishes a real pack-wide load step from a single noisy or failing channel.

5.3 Pack analytics

Pack SoC is the mean of per-cell SoC, where cell SoC is filtered voltage divided by 3.20 V. Imbalance is (strongest minus weakest) in mV. The trend is the change in imbalance per sample scaled to mV/s. The adaptive imbalance threshold depends on pack SoC:

Pack SoC	Threshold (mV)
below 20 %	180
20 to 30 %	150
30 to 40 %	130
40 to 50 %	110
50 to 70 %	100
70 to 80 %	110
80 to 90 %	130
90 % and above	150

5.4 Protection state machine

State	Relay	LEDs	Buzzer	Entry condition
NORMAL	Closed	Green	Off	Boot, or recovery dwell complete, or imbalance at or below 60 % of threshold for 1 s from WARNING
WARNING	Closed	Yellow	Off	Imbalance above 80 % of threshold for 1 s
FAULT	Open	Red	300 ms pulse every 2 s	Any sensor fault immediately, or imbalance above threshold for 0.8 s
RECOVERY	Open	Yellow/green alternating (250 ms)	Off	No sensor fault and imbalance at or below 60 % of threshold for 0.8 s

RECOVERY returns to NORMAL only after an 8 s dwell with no fault; any sensor fault or imbalance above 60 % of threshold during the dwell sends it back to FAULT. The relay therefore stays open for a minimum of about 8.8 s after the fault condition clears. A separate executive state is exposed to the dashboard: NORMAL, DEGRADED (WARNING), FAILSAFE (FAULT and RECOVERY). SHUTDOWN is reserved and never entered.

5.5 LCD engine

The LCD is driven from a 16x2 character shadow buffer. Every 100 ms only characters that differ from what is on the display are written, which removes the flicker caused by lcd.clear(). Three pages (battery status, system state, strongest/weakest cell) rotate every 3 s. In FAULT the page rotation stops and a fixed override screen shows the imbalance and threshold.

5.6 Cloud telemetry

Telemetry is event-driven rather than periodic. A snapshot is created when any of these changes: protection state, relay position, sensor-fault mask, strongest or weakest cell, Wi-Fi or cloud status, any cell by 30 mV or more, RSSI by 8 dBm or more, or every 10 s as a heartbeat.

A snapshot is sent one datastream per 150 ms tick (26 fields, about 3.9 s per snapshot) to stay below Blynk's write-rate limit. V14 (sequence number) is written last and acts as a snapshot-complete marker. If the cloud is unavailable, or older events are still pending, new snapshots go into a 20-entry circular FIFO. On reconnection they are replayed oldest-first at one per 500 ms. If the queue is full the newest event is dropped and counted in V16. Blynk events bms_transition, bms_fault, bms_warning and bms_recovery are logged on state changes while connected.

Pin	Content	Pin	Content
V0 to V3	Cell 1 to 4 voltage	V14	Snapshot sequence (complete marker)

Pin	Content	Pin	Content
V4 / V5	Weakest cell number / voltage	V15	Cloud connected
V6 / V7	Strongest cell number / voltage	V16	Dropped events
V8	Relay closed (1/0)	V17	Pack SoC %
V9	Fault state (0 to 3)	V18	Imbalance mV
V10	RSSI dBm	V19	Adaptive threshold mV
V11	Wi-Fi connected	V20	Risk score
V12	Queue depth	V21	Uptime s
V13	Replayed flag	V22	Total faults
V23	System state code	V24 / V25	Recommendation / state text

Demo commands over the serial monitor: `o` forces a simulated cloud outage, `r` restores it.

5.7 Risk score and recommendation

Risk (0 to 100) = 0.40 x imbalance risk + 0.20 x trend risk + 0.25 x fault-history risk + 0.15 x low-SoC risk, where imbalance risk is imbalance as a percentage of threshold, trend risk is 10 points per mV/s of rising imbalance, fault risk is 20 points per fault in the last 24 h, and SoC risk rises linearly below 30 % SoC. The recommendation text follows a priority order: sensor fault, FAULT, RECOVERY, WARNING, Wi-Fi lost, weak RSSI, rising trend, risk of 60 or more, otherwise normal. The weights are an initial engineering scale and have not been calibrated against real cells.

6. Test Method and Datasets

Two CSV datasets accompany this report. Their expected values were generated by a Python model that re-implements the firmware algorithms step for step (5-sample median, health checks, correlation rule, adaptive threshold, debounce timers, risk score). They have not yet been compared with the firmware running in Wokwi, so the first run of each dataset is itself a verification of both model and firmware.

Manual dataset, bms_manual_test_scenario.csv. Fifteen static scenarios (M01 to M15) to be set with the potentiometers. Each row gives the target cell voltages, the potentiometer slider percentage, the equivalent raw counts, and the expected SoC, imbalance, threshold, final state, relay, LED, LCD page-1 text, risk score and recommendation. The time-to-state column assumes the pots start at a balanced 3.00 V and are changed at one instant.

Timeline dataset, bms_timeline_dataset.csv. Twelve time series (about 1,200 rows) at 5 Hz giving raw ADC counts per cell per sample, with expected filtered voltages, SoC, imbalance, threshold, state, relay, LED, fault mask, transient-suppression flag, system state, fault count, risk and recommendation. These are meant for automated injection (for example a Wokwi automation scenario, or a temporary firmware shim that replaces analogRead).

Common-mode dither. Most timelines add the same pattern of 0, +2, -2, +3, -1, +1, -3, +2 counts to all four cells in each sample. This keeps every sample-to-sample change above the 1-count frozen limit without disturbing imbalance, so the risk trend term stays at zero. Boundary and frozen-sensor timelines use no dither because they need exact counts.

Timeline	Stimulus	Expected outcome
TL01 lifecycle	10 s balanced, 10 s warning set, 10 s fault set, 20 s balanced	NORMAL, WARNING at 11.6 s, FAULT at 21.4 s, RECOVERY at 31.4 s, NORMAL at 39.4 s
TL02 re-fault	Fault, balanced, fault again inside the dwell	FAULT 6.4 s, RECOVERY 11.4 s, FAULT 16.6 s, RECOVERY 20.4 s, NORMAL 28.4 s
TL03 open tap	Cell 3 to 0.10 V (raw 124)	FAULT on the first sample after the change (5.2 s), RECOVERY 11.4 s, NORMAL 19.4 s
TL04 over-range	Cell 2 to 3.25 V (raw 4034)	Same timing as TL03
TL05a to d	All cells at raw 248, 236, 3959, 3971	248 and 3959 stay NORMAL; 236 and 3971 give FAULT on the first sample
TL06 frozen	Constant raw value for 35 s	FAULT at 30.2 s (sample 151)
TL07 transient	All cells 3.00 V to 2.20 V in one step	No fault; suppression flag set on the step sample
TL08 spike	One-sample 1.5 V spike on cell 1	Rejected by the median filter; stays NORMAL
TL09 oscillation	Cell 1 alternates 3.0 V and 1.5 V each sample	Sensor FAULT at 5.6 s, on the third consecutive jump

The manual dataset covers the eight SoC threshold bands (M05 to M11, plus M01), warning and fault at high and mid SoC (M03, M04, M12, M13), and the two hardware-fault cases (M14, M15). In the settled state the risk score is 0.0 for a balanced pack at 40 % SoC or above, 26.6 for M02, about 37 for M03, 45 for the fault cases that have no low-SoC or trend contribution, and 7.2 for a balanced pack at 15.6 % SoC.

Tolerances when comparing results. Voltage within 3 mV, imbalance within 3 mV, risk within 1.0, state-change times within one sample (200 ms) because real loop timing is not an exact multiple of 200 ms, and the dataset time axis starts at the first sample rather than at boot.

7. Future Work

1. Move telemetry and network handling to a separate task so the protection loop is never blocked.
2. Add current and temperature sensing, and use them in the threshold and risk models.
3. Replace the linear SoC with a cell-chemistry lookup table and add coulomb counting.
4. Add relay feedback sensing and a hardware watchdog.

5. Persist the fault history across resets in non-volatile storage.
6. Add a physical reset or acknowledge input for latched-fault operation, and implement the reserved SHUTDOWN policy.
7. Calibrate the risk weights against recorded data.

8. Conclusion

The design meets its core objectives in simulation: reusable analytics, robust sensor validation, a debounced four-state protection sequence with a conservative recovery dwell, a flicker-free local display, and an ordered, queue-backed cloud telemetry path. Before the report's pass criteria can be signed off, the buzzer wiring (F1) should be corrected, the frozen-sensor behaviour in Wokwi (F2) confirmed, and the two datasets run against the firmware. The blocking reconnect (F3) is the main architectural risk to address next.

Appendix A. Key Parameters

Parameter	Value
Cells	4 (`NUM_CELLS`)
Sample interval	200 ms (5 Hz)
Median window	5 samples
Valid cell window	0.20 V to 3.20 V (raw 248 to 3970)
Frozen limit	150 samples (30 s), epsilon 1 count
Jump limit	800 counts, 3 confirmations
Warning / fault debounce	1000 ms / 800 ms
Warning entry / exit	80 % / 60 % of threshold
Recovery dwell	8000 ms
Buzzer pulse / re-beep	300 ms / 2000 ms
LCD refresh / page rotate	100 ms / 3000 ms
Event queue	20 entries, replay every 500 ms
Field pacing	150 ms per datastream, 26 fields
Cell-change / RSSI event delta	30 mV / 8 dBm
Heartbeat snapshot	10 s